A Hierarchical Bayesian Framework for Self-Consistent Sea Level Rise Forecasting: From Global Constraints to Local Projections

Framework Document — Working Draft

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Abstract

No single approach to sea level rise (SLR) projection is sufficient. Process models lack systematic parameter exploration and diverge from observations; purely statistical models cannot resolve component-specific dynamics; expert assessments lack formal self-consistency. We outline a hierarchical Bayesian framework that fuses observational constraints, statistical models, and process-model structural information into a single, self-consistent probabilistic forecast. The framework operates across four levels: (0) global temperature forcing, (1) total GMSL constrained by the observational record, (2) component-level decomposition with component-specific data and models, and (3) deep-uncertainty modules for ice-sheet processes. At each level, posteriors from one analysis become informed priors for the next, and a global budget constraint ensures that components sum to the observed total. Forecast uncertainty is decomposed by horizon: at short lead times (<15 years), observational constraints dominate; at intermediate horizons (15–60 years), physics-informed component models under diverging forcing scenarios provide the primary structure; at long horizons (>60 years), deep-uncertainty modules for Antarctic ice-sheet processes and scenario divergence dominate the projection envelope. The appropriate forecasting methodology and the dominant uncertainty source both change with horizon, and the framework makes this transition explicit and quantifiable. The framework is designed to be built incrementally: it is self-consistent at every stage of completeness, with unresolved components absorbed into a residual term.

1 Motivation

Current approaches to SLR projection fall into three broad categories, each with fundamental limitations:

1. **Process models** (e.g., ISMIP6, GlacierMIP): Simulate ice-sheet and glacier dynamics from first principles, but lack systematic parameter exploration, use heuristic parameterisations for key processes (calving, sub-shelf melting), and produce ensembles that immediately diverge from observations after initialisation (**?**). Under 4°C of warming, IPCC AR6 process-model projections show less SLR by 2100 than a simple extrapolation of observed satellite-era trends — a physically implausible result that reflects structural model deficiencies rather than true uncertainty.

2. **Semi-empirical / statistical models** (e.g., Vermeer & Rahmstorf 2009, this work): Calibrate a rate–temperature relationship against observations, providing strong constraints on the historical period but limited ability to resolve component-specific dynamics or capture nonlinear threshold behaviour (e.g., marine ice-sheet instability).

3. **Expert elicitation and structured assessments** (e.g., IPCC AR6, Bamber et al. 2019): Synthesise multiple lines of evidence but lack formal probabilistic self-consistency between components and total.

The framework described here aims to combine the strengths of all three approaches while providing formal self-consistency through Bayesian hierarchical modelling. The central principle is:

> *Observations constrain the total; component models and data inform the decomposition; the budget ensures consistency; deep-uncertainty modules handle what observations alone cannot resolve.*

1.1 Multi-timescale forecasting

A second organising principle is that no single forecasting methodology is adequate across all time horizons. At short lead times ($\tau < 15$ years), the system trajectory is tightly constrained by its current state and the rate of recent acceleration; observationally calibrated rate models outperform process models, which carry substantial initialisation errors. At intermediate horizons ($15 < \tau < 60$ years), the forcing scenarios diverge, component models extrapolate beyond their calibration domain, and the structural uncertainty grows; the appropriate methodology is physics-informed projection with growing model-inadequacy terms. At long horizons ($\tau > 60$ years), deep uncertainty in ice-sheet dynamics and scenario divergence dominate the projection envelope, and the credibility basis shifts from empirical validation to physical plausibility, conservation laws, and expert assessment.

The framework addresses this by integrating observation-constrained, physics-informed, and deep-uncertainty methods with horizon-appropriate weighting. Rather than prescribing different models at different horizons, the framework uses a single joint model whose variance decomposition (section 5) reveals which uncertainty source dominates at each horizon. The transitions between regimes are not set by fiat — they emerge from the posterior as the relative magnitudes of parameter uncertainty, scenario uncertainty, model inadequacy, and deep uncertainty shift with lead time. This makes the framework's own credibility as a function of horizon transparent and quantifiable (section 5.6).

2 Framework Architecture

The framework operates across four levels, connected by Bayesian conditioning and a budget constraint (fig. 1).

2.1 Level 0: External forcing

Temperature forcing $T(t)$ is treated as an external input, drawn from CMIP6 / IPCC assessed ranges for each SSP scenario. The across-scenario spread contributes $\sigma^2_{\text{scenario}}$ to the total projection uncertainty. Temperature is not a free parameter of the SLR model; it is conditioned upon.

2.2 Level 1: Global GMSL constraint

The polynomial rate–temperature model

$$\text{rate}(t) = a\,T(t)^2 + b\,T(t) + c, \qquad H(t) = H_0 + \int_0^t \text{rate}(\tau)\,\mathrm{d}\tau \tag{1}$$

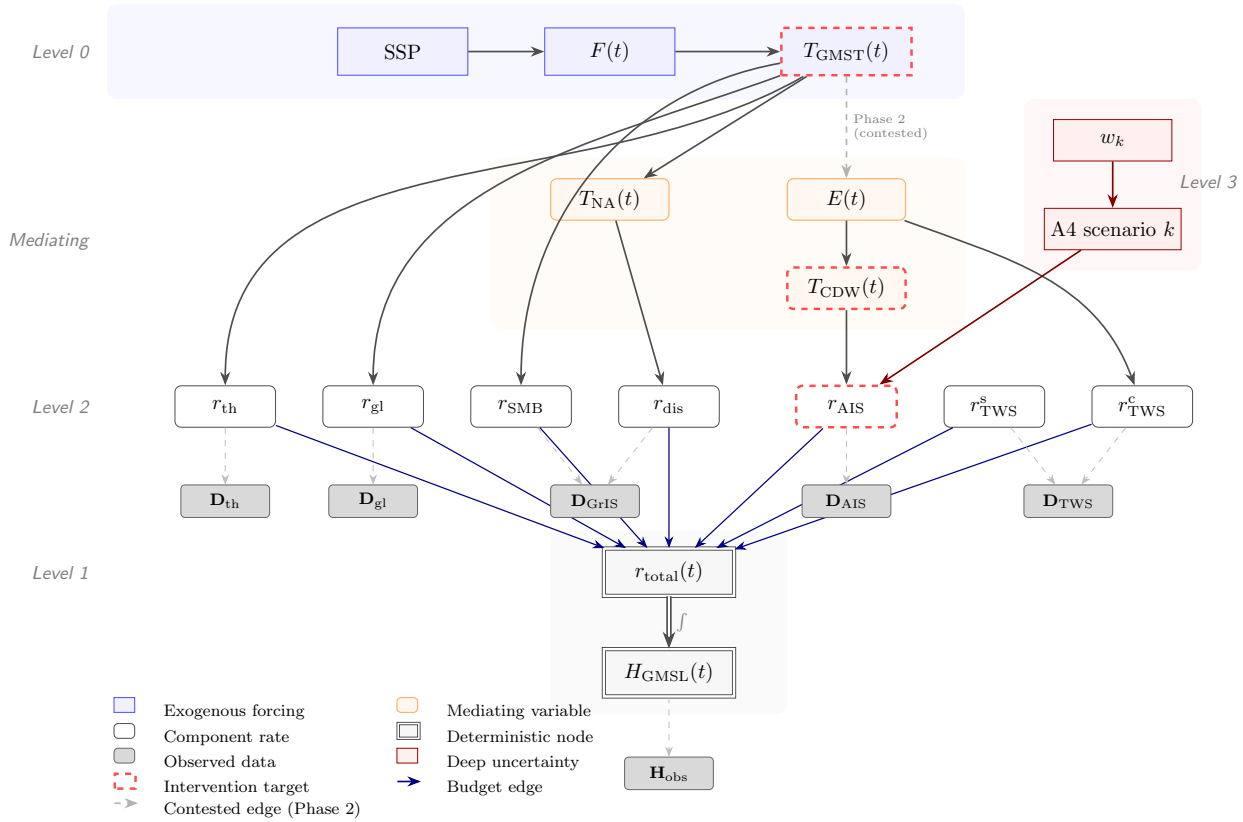

Figure 1: Directed acyclic graph of the hierarchical framework. Nodes are grouped by level: Level 0 (exogenous forcing, blue), Level 2 (component rates, white), Level 1 (budget constraint and GMSL, gray, double-bordered), and Level 3 (deep uncertainty, red). Orange nodes are mediating variables that transmit forcing from Level 0 to specific Level 2 components via physically distinct pathways. Shaded gray nodes are observed data. Dashed red borders mark intervention targets for causal counterfactuals (table 2). Blue edges denote the rate-level budget constraint (soft likelihood, eq. (3)). Dashed gray edges are generative observation models. The dashed edge $T_{\mathrm{GMST}} \to E(t)$ is contested and inactive in Phase 1: ENSO is treated as exogenous (see text). Notation: r_{th} = thermosteric rate; r_{gl} = glacier rate; r_{SMB}, r_{dis} = GrIS surface mass balance and discharge rates; r_{AIS} = AIS rate; $r^{\mathrm{s}}_{\mathrm{TWS}}$, $r^{\mathrm{c}}_{\mathrm{TWS}}$ = TWS secular and climate-driven rates; $E(t)$ = ENSO index; T_{NA} = North Atlantic SST proxy; T_{CDW} = Circumpolar Deep Water temperature proxy.

is calibrated against the full ∼119-year GMSL record using a Bayesian level-space approach (companion document). An Exponential (PC) prior on a provides genuine shrinkage toward the order-1 model, letting the data determine whether quadratic acceleration is supported. A rate-and-state extension with relaxation time τ captures lagged response from slow ocean and ice-sheet processes.

The Level 1 posterior provides:

- A calibrated total rate at any temperature, with uncertainty;
- A total GMSL projection envelope $\sigma^2_{\text{constrained}}$;
- A *budget*: the sum of all component rates must equal the Level 1 total within posterior uncertainty.

2.3 Level 2: Component decomposition

The total rate is decomposed into contributions from distinct physical processes:

$$\text{rate}_{\text{total}}(t) = \text{rate}_{\text{thermo}}(t) + \text{rate}_{\text{glaciers}}(t) + \text{rate}_{\text{GrIS}}(t) + \text{rate}_{\text{AIS}}(t) + \text{rate}_{\text{TWS}}(t) \tag{2}$$

Each component defines its own rate model with its own physically appropriate forcing variable(s) (table 3). The components are not required to share a common polynomial-in-T parameterisation; the only structural requirement is that their rates sum to the observed total.

The budget constraint enters the joint model as a *soft likelihood* rather than a deterministic parameter identity. At each observation time t_j in the calibration period:

$$\text{rate}_{\text{total}}(t_j) - \sum_{i=1}^{K} \text{rate}_i(t_j) \sim \mathcal{N}\big(0,\ \sigma^2_{\text{budget}}(t_j)\big) \tag{3}$$

where

$$\sigma^2_{\text{budget}}(t_j) = \sigma^2_{\text{obs,total}}(t_j) + \sum_{i=1}^{K} \sigma^2_{\text{extra},i} \tag{4}$$

incorporates both the measurement uncertainty in the total GMSL rate and the component-level model inadequacy terms $\sigma_{\text{extra},i}$ (see section 5). In the limit $\sigma_{\text{budget}} \to 0$, this recovers the deterministic budget; allowing slight imbalance prevents numerical pathologies from degenerate posteriors while honestly reflecting that no finite set of component models perfectly accounts for every physical process contributing to the total.

This soft-constraint formulation follows the approach to constraints in Bayesian calibration problems described by **?** and **?**. The budget acts as both a *regulariser* (preventing components from drifting to implausible values) and a *cross-component constraint* (well-observed components tighten the residual and thereby constrain poorly observed ones). Critically, because the constraint operates at every time step rather than through a small number of shared polynomial parameters, the regularisation is stronger: it penalises component models that deviate from the total *anywhere* in the calibration period, not only through their integrated polynomial behaviour.

2.4 Level 3: Deep-uncertainty modules

For components where observational constraints are insufficient to characterise the tails of the distribution — principally the Antarctic Ice Sheet — structured deep-uncertainty modules extend the projection envelope. The A4 scenario-mixture framework (companion document) provides a physically motivated, discrete mixture of ice-sheet futures:

- S1: Status quo (no marine ice-sheet instability);
- S2: Marine ice-sheet instability (MISI);
- S3: MISI with amplifying feedbacks;
- S4: MISI with marine ice-cliff instability (MICI).

Each scenario has its own within-scenario distribution; the mixture weights encode expert assessment of scenario plausibility. The resulting deep-uncertainty contribution enters the total projection variance through the nested law-of-total-variance decomposition (section 5). The approximation that the A4 contribution is independent of the Level 1 calibration uncertainty (Assumption V1 in section 5.4) requires that the A4 scenarios be consistent with the historical budget residual; this condition is tested empirically.

2.5 Formal DAG specification

The DAG in fig. 1 encodes the causal assumptions of the framework. This subsection specifies all nodes, edges, implied conditional independence relations, and intervention targets. The specification serves three purposes: (1) it makes the conditional independence structure of the joint posterior (eq. (7)) explicit and verifiable; (2) it defines which nodes support do-operations for counterfactual analysis; and (3) it provides the formal object against which any future extension of the framework must be checked for consistency.

2.5.1 Node set

2.5.2 Edge set

The directed edges encode generative (causal) relationships. Each edge $A \to B$ asserts that B is generated by a mechanism that takes A as input; the conditional distribution $p(B \mid \mathrm{pa}(B))$ is part of the joint model specification.

Forcing chain (Level 0). SSP $\to F(t) \to T_{\mathrm{GMST}}(t)$. Radiative forcing is determined by the SSP scenario; GMST responds to forcing via the climate system (treated as a known input, not a free parameter of the SLR model).

GMST $\to$ component rates (direct). $T_{\mathrm{GMST}} \to r_{\mathrm{th}}$, $T_{\mathrm{GMST}} \to r_{\mathrm{gl}}$, $T_{\mathrm{GMST}} \to r_{\mathrm{SMB}}$. These components respond directly to global mean temperature through thermodynamic mechanisms (thermal expansion, surface energy balance, melt).

GMST $\to$ mediating variables. $T_{\mathrm{GMST}} \to T_{\mathrm{NA}}$: GMST influences North Atlantic subsurface temperature, mediated by Atlantic Meridional Overturning Circulation and regional ocean dynamics. $T_{\mathrm{GMST}} \to E(t)$: warming may shift ENSO statistics (**?**), but the hypothesis is not robustly supported by CMIP6 multi-model ensembles, which show no agreement on the sign or magnitude of the ENSO response to warming. **Phase 1 treatment**: ENSO is exogenous. The $T_{\mathrm{GMST}} \to E(t)$ edge is *inactive*: ENSO statistics are drawn from the historical record or from a specified scenario, not generated endogenously by the model. This means that do-interventions on $F(t)$ (SAI, CDR) do *not* propagate to r_{AIS} or $r^{\mathrm{c}}_{\mathrm{TWS}}$ via ENSO in Phase 1. The edge will be activated in Phase 2 with a mandatory sensitivity analysis comparing counterfactual projections with and without the edge active (see fig. 1, where the edge is shown dashed).

Teleconnection edges. $T_{\mathrm{NA}} \to r_{\mathrm{dis}}$: ocean thermal forcing on marine-terminating Greenland outlet glaciers (**?**). $E(t) \to T_{\mathrm{CDW}}$: ENSO modulates CDW delivery to the Antarctic continental shelf via the Pacific–South American teleconnection (**??**). $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$: sub-shelf basal melt rates respond to CDW temperature. $E(t) \to r^{\mathrm{c}}_{\mathrm{TWS}}$: ENSO drives terrestrial water storage variability through precipitation shifts.

Budget edges (deterministic). $\{r_{\mathrm{th}}, r_{\mathrm{gl}}, r_{\mathrm{SMB}}, r_{\mathrm{dis}}, r_{\mathrm{AIS}}, r^{\mathrm{s}}_{\mathrm{TWS}}, r^{\mathrm{c}}_{\mathrm{TWS}}\} \to r_{\mathrm{total}}$: the total rate is the deterministic sum of component rates. $r_{\mathrm{total}} \to H_{\mathrm{GMSL}}$: GMSL is the time integral of the total rate.

Deep-uncertainty edges (Level 3). $w_k \to k$: the A4 scenario index is drawn from the categorical distribution defined by the mixture weights. $k \to r_{\mathrm{AIS}}$: the A4 scenario determines the AIS rate distribution during projection. *Future edge (Phase 2)*: the Glen–Nye flow law exponent n (sampled from a prior on $[3, 4]$) will modify the within-scenario AIS rate distributions via a rheology correction following the analysis of **?**. The edge $n \to r_{\mathrm{AIS}}$ couples the A4 projection to ice rheology uncertainty, which is currently treated as an implicit structural assumption within each A4 scenario rather than as a sampled parameter.

Observation edges (generative). $H_{\mathrm{GMSL}} \to \mathbf{H}_{\mathrm{obs}}$: the GMSL observation model (measurement noise + datum uncertainty). $r_i \to \mathbf{D}_i$ for each component with direct observations: the component-specific measurement models. Note that $\mathbf{D}_{\mathrm{GrIS}}$ is generated jointly by r_{SMB} and r_{dis}, and $\mathbf{D}_{\mathrm{TWS}}$ is generated jointly by $r^{\mathrm{s}}_{\mathrm{TWS}}$ and $r^{\mathrm{c}}_{\mathrm{TWS}}$.

2.5.3 Conditional independence structure

The DAG implies the following conditional independence (CI) relations. Each is stated with its physical justification and the conditions under which it could be violated.

Critical distinction: generative vs. inferential dependence. The CI relations below describe the *generative* model — the data-generating process assumed by the framework. During *inference*, the dependence structure is different, and the difference is central to how the framework operates. The total rate node r_{total} is a *collider*: all seven component rates feed into it via the budget sum. In the generative model, where r_{total} is unobserved, the component rates are d-separated from each other given their respective forcings (see CI-1–CI-3 below). However, during inference, r_{total} is effectively conditioned upon, because the GMSL likelihood $p(\mathbf{H}_{\mathrm{obs}} \mid \ldots)$ constrains $H_{\mathrm{GMSL}}(t) = \int r_{\mathrm{total}}(\tau)\,\mathrm{d}\tau$. By the rules of d-separation, conditioning on a collider *opens* paths between its parents that were previously blocked. Concretely: once the total rate is constrained by the GMSL observations, learning that one component rate is large forces the remaining components to be collectively smaller, and vice versa.

This collider-induced coupling is not a pathology — it is the *mechanism* by which the budget constraint creates cross-component information flow. It is the formal graphical-model statement of the "top-down constraint" described in section 4: well-constrained components (e.g., thermosteric) tighten the budget residual and thereby indirectly constrain poorly observed components (e.g., AIS). The CI relations below hold in the generative model but are systematically violated during inference. Where this violation is substantive, it is noted explicitly.

1. **CI-1**: $r_{\mathrm{th}} \perp\!\!\!\perp r_{\mathrm{AIS}} \mid T_{\mathrm{GMST}}, E(t)$. The thermosteric rate is independent of the AIS rate given GMST and ENSO state. *Justification*: the only coupling is through shared forcing; there

is no direct thermosteric $\rightarrow$ AIS or AIS $\rightarrow$ thermosteric mechanism. *Potential violation*: freshwater input from AIS mass loss could modify ocean stratification and thereby affect thermosteric expansion at regional scales. This is a second-order effect that is negligible at the global level over the calibration period but could become relevant under high-end AIS loss scenarios. *Inferential violation*: conditioning on r_{total} (via $\mathbf{H}_{\text{obs}}$) opens the collider path $r_{\text{th}} \rightarrow r_{\text{total}} \leftarrow r_{\text{AIS}}$. During inference, the thermosteric and AIS rates become negatively correlated: a posterior draw in which the thermosteric rate is higher than expected leaves less budget residual for AIS, and vice versa. This is precisely the mechanism by which the well-constrained Argo-era thermosteric estimate tightens the AIS contribution.

2. **CI-2**: $r^{\text{c}}_{\text{TWS}} \perp\!\!\!\perp r_{\text{AIS}} \mid E(t)$. The climate-driven TWS rate and the AIS rate are conditionally independent given ENSO state. *Justification*: ENSO is the sole common driver; no direct TWS $\rightarrow$ AIS or AIS $\rightarrow$ TWS pathway exists. *Potential violation*: if additional unmodelled climate modes (e.g., Indian Ocean Dipole, SAM) simultaneously affect both TWS and CDW delivery, the CI relation is approximate. This is mitigated by the fact that ENSO is the dominant mode on the relevant timescales. *Inferential violation*: same collider mechanism as CI-1. Conditioning on the budget total induces posterior correlation between $r^{\text{c}}_{\text{TWS}}$ and r_{AIS} even when ENSO is conditioned upon. The induced correlation is weaker than for CI-1 because $r^{\text{c}}_{\text{TWS}}$ has smaller magnitude, but it is structurally present in the posterior.

3. **CI-3**: $r_{\text{gl}} \perp\!\!\!\perp r_{\text{dis}} \mid T_{\text{GMST}}$. Glacier rate is independent of GrIS discharge rate given GMST. *Justification*: glacier melt responds to atmospheric temperature; GrIS discharge responds to North Atlantic ocean temperature. Both are driven by GMST but through different mechanisms with no direct cross-pathway. *Potential violation*: high-latitude glacier mass loss (e.g., Svalbard, Iceland) may share regional ocean forcing with Greenland marine-terminating glaciers. This is a regional effect absorbed by the model-inadequacy terms.

4. **CI-4**: $\mathbf{D}_i \perp\!\!\!\perp \mathbf{D}_j \mid r_i, r_j$ for $i \neq j$ when the datasets are instrumentally independent. *Justification*: given the true component rates, the measurement errors are independent if the instruments are distinct. *Potential violation*: GRACE provides mass-change estimates for glaciers, GrIS, AIS, and TWS simultaneously; if the GRACE processing introduces correlated errors across these products, CI-4 is violated for GRACE-derived datasets. This must be handled by a correlated error model for GRACE components (deferred to Phase 2 unless cross-product GRACE correlations are large enough to affect the posterior).

5. **CI-5**: $r^{\text{s}}_{\text{TWS}} \perp\!\!\!\perp r_i \mid$ nothing, for all $i \neq \text{TWS}^{\text{s}}$. The secular TWS rate has no parents in the DAG other than its own parameters; it is marginally independent of all climate-driven rates. *Justification*: dam impoundment and groundwater extraction are driven by human decisions, not by climate forcing. This is the strongest CI assumption in the framework and is well-supported physically.

These CI relations should be tested empirically during implementation. CI-1 and CI-2 can be checked by examining residual correlations between components after conditioning on the relevant forcing variables. CI-4 requires assessment of GRACE inter-product error correlations. The collider-induced inferential violations of CI-1 through CI-3 and CI-5 are not testable in the usual sense — they are structural consequences of conditioning on the budget constraint and are present by construction. They can, however, be *quantified* by computing pairwise posterior correlations $\text{Cor}[r_i(t), r_j(t) \mid \mathbf{H}_{\text{obs}}, \mathbf{D}]$ and comparing to the generative prior correlations. The difference measures the strength of the top-down budget regularisation at each time step. Large inferential

correlations indicate times where the budget is doing substantial work to constrain the decomposition; small correlations indicate times where the component-specific data are sufficient on their own.

2.5.4 Intervention targets

For causal counterfactual analysis — specifically, for evaluating proposed glacier stabilisation interventions under the Arête Glacier Initiative — certain nodes must be manipulable via Pearl's do-operator (**?**). A do-operation $do(X = x)$ replaces the natural generative mechanism for X (i.e., deletes all incoming edges to X) and fixes X at the specified value. The post-intervention distribution is obtained by graph surgery on the DAG.

The DAG structure ensures that counterfactual queries are well-defined: for any intervention on any marked node, the post-intervention distribution is uniquely determined by the truncated factorisation (**?**). The key requirement is that the causal mechanisms encoded by the edges are *autonomous* — that is, intervening on one mechanism does not alter others. This autonomy assumption is defensible for the interventions in table 2: stabilising a grounding line does not alter the physics of thermal expansion, and modifying radiative forcing does not alter the dam-impoundment history.

Phase 1 limitation. In Phase 1, the component rate models are statistical (polynomial, linear, or index-based), not mechanistic. This means that counterfactual queries that push the forcing far outside the calibration range will produce extrapolations whose reliability depends on the structural correctness of the statistical model. Phase 2 (emulator-based component models) will address this by replacing statistical rate models with emulators trained on mechanistic simulators, which have more defensible extrapolation behaviour within their domain of physical validity.

Value-of-information queryability. The DAG structure determines which observations reduce uncertainty on which projections. For any hypothetical future observation at node X, the set of downstream nodes whose posterior variance would be reduced can be identified by graph-theoretic computation on the DAG without re-running inference: it is the set of nodes reachable from X in the moral graph induced by the current evidence set. This property supports value-of-information (VoI) analysis for observational investment decisions (Phase 3). The Phase 1 design decisions that preserve this capability are: (i) forward models accept arbitrary (t) evaluation points, not only the observation grid; (ii) the likelihood interface supports appending hypothetical observations by specifying type, time, value, and uncertainty; (iii) MCMC outputs store full parameter vectors and per-observation log-likelihoods to enable importance reweighting under hypothetical data augmentation.

Sequential update interface (Phase 1 design, Phase 2 implementation). The principle that forecasts should be updated as new observations arrive requires formal support. New data arrive annually (GMSL from altimetry, GRACE mass-change estimates, Argo thermosteric profiles), and the framework must incorporate this information without full re-estimation at each update. Three update algorithms are available in order of increasing computational cost: (A) importance reweighting of the stored MCMC sample, (B) sequential Monte Carlo (particle filter with rejuvenation), and (C) full MCMC re-estimation.

Option A is the Phase 1 target. Given the stored posterior sample $\{\boldsymbol{\theta}^{(m)}\}_{m=1}^{M}$ and new data $\mathbf{D}_{\text{new}}$, the importance weights are $w^{(m)} \propto p(\mathbf{D}_{\text{new}} \mid \boldsymbol{\theta}^{(m)})$. The effective sample size $n_{\text{eff}} = (\sum w^{(m)})^2 / \sum (w^{(m)})^2$ measures how much the new data have changed the posterior. If $n_{\text{eff}}/M <$

0.5, the reweighted sample is unreliable and full re-estimation is triggered. This degrades after ∼3 years of cumulative updates, at which point Option C is required. Option B (SMC) is the Phase 2 implementation target, as it avoids degeneracy indefinitely but requires more complex implementation.

The following Phase 1 design requirements ensure that sequential updating is retrofittable in Phase 2 without restructuring the MCMC output format:

1. **Full MCMC storage:** Each posterior sample stores the complete parameter vector $\boldsymbol{\theta}^{(m)}$ and the per-observation log-likelihood vector $\ell^{(m)}(t_j)$ for $j = 1, \ldots, N_t$.

2. **Modular likelihood:** The likelihood function accepts an arbitrary set of observation times, values, and uncertainties. Adding a new observation is a single function call that extends the existing observation arrays.

3. **Arbitrary-time evaluation:** Component rate models accept any time t, not only the calibration grid, so that predictions can be evaluated at future observation times.

4. **Forecast evaluation function:** Given a posterior sample and a new observation, compute the posterior predictive at the observation time, the coverage indicator, and the CRPS (section 6). This function is the building block for both the hindcast protocol and real-time forecast validation.

3 Component Models and Informed Priors

Each component analysis provides a marginal posterior on its own parameters, which serves as an *informed prior* when incorporated into the joint hierarchical model. The strength of the prior reflects the quality and length of the component-specific observational record. Because the budget operates at the rate level (eq. (3)), components are not required to share a parameterisation; each uses the functional form and forcing variables most appropriate to its physics (table 3).

3.1 Thermosteric expansion

Data. Argo float array (2005–present, global coverage below 2000 m); ocean reanalysis products (1950–present, declining coverage with depth and time); XBT profiles (1960s–present, upper 700 m, with known fall-rate biases).

Model. Thermal expansion is well-understood from first principles: $\text{rate}_{\text{thermo}}(T) \approx b_{\text{th}} \cdot T + c_{\text{th}}$. The response is expected to be nearly linear in global mean temperature because the thermal expansion coefficient varies slowly over the relevant temperature range. A quadratic term $a_{\text{th}}T^2$ is available but expected to be small; an Exponential (PC) prior on a_{th} provides shrinkage toward the linear model.

Prior contribution. Because thermal expansion is the best-constrained component, the prior on b_{th} will be tight: approximately $\text{Normal}(\mu_{\text{th}}, \sigma_{\text{th}})$ with σ_{th} determined by the Argo-era uncertainty. This tight prior propagates through the rate-level budget to constrain the residual available to other components at every calibration time step.

Horizon-dependent inadequacy. The thermosteric growth function is $g_{\mathrm{th}}(\tau) = 1 - \exp(-\tau/\tau_{\mathrm{th,struct}})$ with $\tau_{\mathrm{th,struct}} \sim 150$ yr. Growth is slow because thermal expansion physics is well-understood: the thermal expansion coefficient has finite variability, and the main risk is deep-ocean heat uptake efficiency changing under novel forcing. The exponential saturation reflects the bounded nature of thermosteric model error (table 4).

3.2 Glaciers and ice caps

Data. Randolph Glacier Inventory (RGI); geodetic mass balance estimates; GRACE/GRACE-FO (2002–present); regional temperature and precipitation records.

Model. Glacier mass balance responds to both temperature (melt) and precipitation (accumulation). The temperature sensitivity may be nonlinear because: (a) small glaciers disappear entirely, reducing the total glacier area; (b) the elevation–temperature feedback amplifies warming at glacier surfaces. A model of the form $\mathrm{rate}_{\mathrm{glaciers}}(T) = a_{\mathrm{gl}} \cdot T^2 + b_{\mathrm{gl}} \cdot T + c_{\mathrm{gl}}$ is appropriate, with $a_{\mathrm{gl}} > 0$ reflecting the accelerating loss as glacier area shrinks.

Prior contribution. Moderate constraint on b_{gl} from the geodetic record; weak constraint on a_{gl} from the relatively short satellite era. The key question is whether the observed glacier acceleration is separable from ice-sheet acceleration in the GRACE record.

Horizon-dependent inadequacy. The glacier growth function is $g_{\mathrm{gl}}(\tau) = \min\big(1,\ [\tau/\tau_{\mathrm{gl,struct}}]^{1/2}\big)$ with $\tau_{\mathrm{gl,struct}} \sim 70$ yr. The square-root form reflects a random-walk analogy for structural misspecification: the quadratic-in-T model extrapolates poorly in the intermediate regime ($\sim$30–80 yr) where glacier area-loss feedback is nonlinear, but volume-depletion saturation provides a natural ceiling that bounds the structural error (table 4).

3.3 Greenland Ice Sheet

Data. GRACE/GRACE-FO; SMB models (RACMO, MAR); ice discharge estimates from satellite velocity and thickness.

Model. Greenland mass loss has two distinct channels: surface mass balance (SMB, strongly temperature-dependent) and ice discharge (driven primarily by ocean–ice interaction at marine-terminating outlet glaciers). A two-component model:

$$\mathrm{rate}_{\mathrm{GrIS}}(t) = \underbrace{b_{\mathrm{SMB}} \cdot T(t) + c_{\mathrm{SMB}}}_{\text{SMB (GMST-forced)}} + \underbrace{\beta_{\mathrm{dis}} \cdot T_{\mathrm{NA}}(t) + c_{\mathrm{dis}}}_{\text{Discharge (ocean-forced)}} \tag{5}$$

where $T_{\mathrm{NA}}(t)$ is a North Atlantic ocean temperature proxy. The discharge term is driven by ocean thermal forcing on marine-terminating outlet glaciers, not by GMST directly; the relationship between GMST and North Atlantic subsurface temperature is mediated by Atlantic meridional overturning and regional ocean dynamics. During the pre-Argo era, T_{NA} may need to be treated as a latent variable or parameterised as a function of GMST with substantial uncertainty.

Prior contribution. The SMB component provides an informed prior on b_{SMB} from regional climate models (RACMO, MAR); the discharge component contributes through β_{dis} and c_{dis}, constrained by satellite-era velocity and thickness observations.

Horizon-dependent inadequacy. The two GrIS sub-components have distinct growth functions. The SMB growth function is $g_{\mathrm{SMB}}(\tau) = 1 - \exp(-\tau/\tau_{\mathrm{SMB,struct}})$ with $\tau_{\mathrm{SMB,struct}} \sim 100$ yr, reflecting that SMB is well-constrained by temperature with the main risk being nonlinear surface-albedo feedback and firn-aquifer dynamics under sustained warming. The discharge growth function is $g_{\mathrm{dis}}(\tau) = 1 - \exp(-\tau/\tau_{\mathrm{dis,struct}})$ with $\tau_{\mathrm{dis,struct}} \sim 40$ yr — faster growth than SMB because discharge dynamics depend on fjord geometry, ice-ocean interaction, and buttressing loss, all of which are poorly constrained by historical data and increasingly important under warming (table 4).

3.4 Antarctic Ice Sheet

Data. GRACE/GRACE-FO; altimetry (CryoSat-2, ICESat-2); InSAR-derived ice velocities; ocean temperature observations on the continental shelf.

Model. The AIS contribution is the least constrained and most consequential for high-end projections. The historical contribution (∼0.3 mm/yr, ∼8–9% of the total GMSL rate) is small relative to the observational uncertainty, and the future trajectory depends on poorly constrained processes (marine ice-sheet instability, ice-cliff failure, sub-shelf melting parameterisation).

In the hierarchical framework, the AIS operates at two levels:

- **Budget-constrained estimate**: The Level 1 total minus the sum of better-constrained components yields a *residual* that bounds the historical AIS contribution. This is an indirect but powerful constraint.
- **Deep-uncertainty extension**: The A4 scenario-mixture module (Level 3) extends the projection envelope beyond what the historical record can constrain.

Prior contribution. The AIS rate is *not* modelled as a polynomial in GMST. Dynamical mass loss is driven by ocean thermal forcing on the Antarctic continental shelf — warm Circumpolar Deep Water (CDW) intrusions modulated by atmospheric teleconnections (Pacific–South American pattern, Southern Annular Mode) and ocean circulation — and the relationship between GMST and CDW temperature at the Amundsen Sea grounding zones is neither linear nor monotonic. During calibration, the AIS contribution is primarily constrained by the budget residual: the Level 1 total minus the sum of all better-constrained components. During projection, the A4 module (Level 3) provides the AIS rate distribution conditioned on scenario. The budget residual provides indirect but powerful constraint: any systematic bias in the AIS rate would manifest as a misfit to the total GMSL record propagated through the budget likelihood (eq. (3)).

Horizon-dependent inadequacy. The AIS component does not use the $g_i(\tau)$ growth-function mechanism. The A4 deep-uncertainty module already provides horizon-dependent uncertainty through its scenario structure: the within-scenario AIS rate distributions widen with lead time, and the scenario weights shift toward higher-risk scenarios as the forcing departs from the historical range. During the calibration period, the AIS model-inadequacy scalar $\sigma_{\mathrm{extra,AIS,0}}$ reflects the budget-residual uncertainty. During projection, the A4 module replaces this scalar entirely. The transition is handled by the forecast regime structure (section 5.6): in Regime 1, the budget-residual constraint dominates; in Regime 3, the A4 module dominates; in Regime 2, both contribute.

3.5 Terrestrial water storage

Data. GRACE/GRACE-FO; dam construction records; groundwater depletion estimates; land hydrology models.

Model. TWS is often treated as a purely non-climatic contribution (dam impoundment, groundwater extraction), contributing only to the background rate c. This is an oversimplification. TWS has both secular and climate-driven components:

$$\mathrm{rate}_{\mathrm{TWS}}(t) = c_{\mathrm{TWS,secular}} + \mathrm{rate}_{\mathrm{TWS,climate}}(t) \tag{6}$$

Climate-driven TWS and the ENSO–WAIS teleconnection. The climate-driven component of TWS is substantial and coupled to modes of climate variability, particularly the El Niño–Southern Oscillation (ENSO). ENSO drives large interannual fluctuations in terrestrial water storage through shifts in precipitation patterns, river discharge, and soil moisture. These fluctuations are visible in both the GRACE TWS record and in GMSL residuals after removing the secular trend.

Critically, ENSO also modulates the delivery of warm Circumpolar Deep Water (CDW) to the Antarctic continental shelf (**??**). During El Niño events, anomalous atmospheric circulation patterns (transmitted via the Pacific–South American teleconnection) weaken the westerlies along the Amundsen Sea coast, altering the thermocline depth and allowing warm CDW to reach the grounding zones of Thwaites and Pine Island glaciers. This drives enhanced basal melting, grounding-line retreat, and — if sustained — potentially irreversible marine ice-sheet instability.

The implication for the hierarchical framework is that TWS and AIS are *not independent components*. The same climate mode (ENSO) that drives short-timescale TWS variability also modulates the ocean forcing that controls WAIS stability on longer timescales. This coupling operates across timescales:

- **Interannual (∼2–7 yr):** ENSO-driven TWS fluctuations and ENSO-driven CDW intrusions are essentially simultaneous, driven by the same atmospheric teleconnection. On these timescales, the TWS and AIS signals may partially cancel in the GMSL budget (land water storage increases during La Niña, when CDW intrusions are weaker).
- **Decadal (∼10–30 yr):** Sustained periods of El Niño–like conditions (e.g., the positive phase of the Interdecadal Pacific Oscillation) can drive cumulative grounding-line retreat that crosses irreversibility thresholds. The TWS signal integrates to near zero over these timescales, but the ice-sheet response is path-dependent and potentially irreversible.
- **Centennial (∼100+ yr):** If anthropogenic warming shifts the mean state of the tropical Pacific toward more frequent or intense El Niño conditions (**?**), the time-mean CDW delivery to the Amundsen Sea increases, loading the dice for MISI even in the absence of extreme individual events.

Implications for the framework. The ENSO–TWS–WAIS coupling has three consequences for the hierarchical model:

1. **Cross-component covariance:** The component decomposition (eq. (2)) must allow for non-zero covariance between $\mathrm{rate}_{\mathrm{TWS}}$ and $\mathrm{rate}_{\mathrm{AIS}}$ at interannual timescales. A diagonal error model (independent components) would underestimate the total uncertainty on short timescales and overestimate it on long timescales.

2. **State-dependent AIS forcing:** The rate-and-state model's relaxation time τ may need to be conditioned on ENSO state or its long-term statistics, rather than treated as a fixed parameter. A regime-dependent τ (fast response during El Niño, slow during La Niña) could capture the asymmetric forcing.

3. **Irreversibility and path dependence:** The A4 scenario-mixture weights (Level 3) should be informed by the projected evolution of ENSO statistics under warming. If climate models project more frequent El Niño conditions, the probability weight on S2–S4 (MISI scenarios) should increase, because the cumulative CDW forcing is more likely to trigger irreversible retreat.

Prior contribution. TWS contributes an informed prior on c_{TWS} from dam and groundwater records. The climate-driven component is parameterised as $\gamma_{\mathrm{TWS}} \cdot E(t)$ where $E(t)$ is an ENSO index, because the TWS–temperature relationship is mediated by ENSO rather than by direct thermodynamic forcing. Forcing TWS into a polynomial-in-T model would create a spurious temperature dependence that contaminates the budget.

Horizon-dependent inadequacy. The two TWS sub-components have distinct treatments. The secular component has $g_{\mathrm{TWS,s}}(\tau) = \tau/\tau_{\mathrm{TWS,struct}}$, reflecting linear growth: the future trajectory of dam construction and groundwater extraction depends on human decisions, and there is no physical bound on how far the future TWS secular rate may deviate from the fitted trend. The scale is set such that $\sigma_{\mathrm{struct,TWS}} \cdot \sqrt{g(80)} \approx 0.5$ mm/yr, an order-of-magnitude bound on TWS rate uncertainty at 2100. The climate-driven component has $g_{\mathrm{TWS,c}}(\tau) = 0$ (no growth): the model $\gamma_{\mathrm{TWS}} \cdot E(t)$ is equally valid at all times because the functional form (linear in ENSO index) does not degrade with forecast lead time — the uncertainty in the TWS climate response lies in the ENSO trajectory itself, which is scenario uncertainty, not model inadequacy (table 4).

3.6 Horizon-dependent model averaging (Phase 2)

In Phase 2, multiple model classes may be available for some components (e.g., for glaciers: the polynomial-in-T model, a GP emulator trained on GlacierMIP output, and a PINN surrogate). The relative skill of these models depends on forecast horizon: the polynomial may be best at short horizons where it is essentially interpolating, while the emulator may be best at long horizons where it captures nonlinear behaviour that the polynomial misses.

Horizon-dependent Bayesian model averaging (BMA) addresses this by assigning model weights that vary with lead time. The component node interface already supports multiple model classes: any model that provides a rate function $r_i(t; \boldsymbol{\theta}_i)$, a prior $p(\boldsymbol{\theta}_i)$, and a model-inadequacy term $\sigma_{\mathrm{extra},i}(\tau)$ can be registered at a given node. Multiple model classes at the same node are handled by augmenting the parameter space with a discrete model indicator $M_i \in \{1, \ldots, J_i\}$. The hindcast protocol (section 6) produces horizon-dependent CRPS for each model class, from which BMA weights at horizon τ can be estimated.

Implementation of horizon-dependent BMA is entirely Phase 2. The Phase 1 requirement is limited to verifying that the node interface can accept multiple model classes without modification.

4 Bayesian Mechanics of the Hierarchy

4.1 Joint model

The full hierarchical model has the following structure. Let $\boldsymbol{\theta}_i$ denote the parameter vector for component i (with component-specific dimension and interpretation; see table 3), and let $\boldsymbol{\theta}_{\text{global}} = (a, b, c, \tau, H_0, \sigma_{\text{extra}})$ denote the Level 1 global parameters. The joint posterior is:

$$
\begin{aligned}
&p(\boldsymbol{\theta}_{\text{global}}, \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K, \boldsymbol{\eta} \mid \mathbf{H}_{\text{obs}}, \mathbf{D}_1, \ldots, \mathbf{D}_K) \\
&\quad \propto \underbrace{p(\mathbf{H}_{\text{obs}} \mid \boldsymbol{\theta}_{\text{global}}, \boldsymbol{\eta})}_{\text{GMSL likelihood}} \cdot \underbrace{\prod_{j=1}^{N_t} p_{\text{budget}}\left(\text{rate}_{\text{total}}(t_j) - \sum_{i=1}^{K} \text{rate}_i(t_j;\, \boldsymbol{\theta}_i) \;\middle|\; \sigma_{\text{budget}}(t_j) \right)}_{\text{rate-level budget constraint}} \\
&\quad \cdot \prod_{i=1}^{K} \underbrace{p(\mathbf{D}_i \mid \boldsymbol{\theta}_i)}_{\text{component likelihoods}} \cdot \prod_{i=1}^{K} \underbrace{p(\boldsymbol{\theta}_i)}_{\text{component priors}} \cdot \underbrace{p(\boldsymbol{\eta})}_{\text{nuisance priors}}
\end{aligned}
\tag{7}
$$

where p_{budget} is the Gaussian soft constraint from eq. (3), $\boldsymbol{\eta}$ collects nuisance parameters, and $\mathbf{D}_i$ is the component-specific dataset. The budget tolerance $\sigma_{\text{budget}}(t_j)$ is given by eq. (4).

Two structural differences from a coefficient-level budget formulation deserve emphasis:

1. The budget is a *likelihood term*, not a deterministic parameter constraint. The component parameters $\boldsymbol{\theta}_i$ are not algebraically linked to the global parameters $\boldsymbol{\theta}_{\text{global}}$; they are coupled through the rate-level budget likelihood, which penalises configurations where the component rates fail to sum to the total at any calibration time step.

2. The component parameter vectors $\boldsymbol{\theta}_i$ have different dimensions and physical interpretations. There is no requirement that each component parameterises its rate as a polynomial in GMST. The AIS rate may be a budget residual or a function of CDW temperature; the TWS climate rate may depend on the ENSO index; the GrIS discharge rate may depend on North Atlantic SST. The only requirement is that each $\text{rate}_i(t;\, \boldsymbol{\theta}_i)$ is evaluable at the calibration time points.

4.2 Information flow

Information flows in both directions through the hierarchy:

Bottom-up (component data → global constraint). Each component likelihood $p(\mathbf{D}_i \mid \boldsymbol{\theta}_i)$ constrains the component rate at times where component-specific observations are available. Through the rate-level budget (eq. (3)), these component-level constraints propagate upward: well-constrained component rates reduce the residual budget available to other components, effectively tightening the total.

Top-down (global likelihood → component posteriors). The GMSL likelihood constrains the total rate $\text{rate}_{\text{total}}(t)$ at every calibration time step. Components that are poorly observed directly (e.g., AIS before GRACE) receive indirect constraint from the budget: if the sum of well-observed component rates accounts for most of the observed total, the budget likelihood penalises any component configuration whose residual deviates significantly from the remainder.

Cross-component (budget → pairwise constraints). Even without direct observations of a component, the budget and the posteriors on other components provide information. For example, if thermosteric + glaciers + GrIS + TWS account for $X \pm \sigma_X$ mm/yr at time t, and the total rate

is $Y \pm \sigma_Y$ mm/yr, then the AIS rate is constrained to $Y - X \pm \sqrt{\sigma_Y^2 + \sigma_X^2}$ — an *observationally derived* estimate of the AIS contribution that requires no ice-sheet model. Because this constraint operates at every time step, it is substantially more informative than a constraint on a few shared polynomial coefficients: the component rates must match the total *everywhere* in the calibration window, not merely on average.

4.3 Staged implementation

The framework is designed to be built incrementally. At any stage, unresolved components are absorbed into a residual:

$$\text{rate}_{\text{residual}}(t) = \text{rate}_{\text{total}}(t) - \sum_{i \in \text{resolved}} \text{rate}_i(t) \tag{8}$$

1. **Stage 0 (current):** Global GMSL constraint only. All components are in the residual. The PC prior on a and the rate-and-state extension provide the forecasting model.
2. **Stage 1:** Add thermosteric decomposition. This is the best-constrained component and provides the strongest budget constraint on the remaining (primarily cryospheric) residual.
3. **Stage 2:** Add glaciers and Greenland. These are moderately well-observed and further tighten the residual, which now primarily reflects AIS + TWS.
4. **Stage 3:** Add TWS (secular and climate-driven components) and the ENSO covariate structure. The residual now isolates the AIS contribution.
5. **Stage 4:** Add AIS with the A4 deep-uncertainty module and the ENSO–CDW coupling. The framework is now complete.

At each stage, the framework is *self-consistent*: the budget holds, the posteriors are valid, and projections can be made. Adding a new component *refines* the projections but does not invalidate the previous stage.

5 Variance Decomposition and Total Uncertainty

5.1 Law of total variance decomposition

The total GMSL projection $H(t)$ depends on three sources of uncertainty: the SSP scenario index S (discrete), the A4 deep-uncertainty scenario index k (discrete), and the Level 1 + Level 2 parameters $\boldsymbol{\theta}$ (continuous). The total variance is given by iterated application of the law of total variance:

$$\text{Var}[H(t)] = \underbrace{E_S[\text{Var}_{\boldsymbol{\theta},k}[H(t) \mid S]]}_{\text{expected within-scenario variance}} + \underbrace{\text{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]}_{\text{across-scenario variance}} \tag{9}$$

The within-scenario variance itself decomposes over the A4 deep-uncertainty structure:

$$\text{Var}_{\boldsymbol{\theta},k}[H(t) \mid S] = \underbrace{E_k[\text{Var}_{\boldsymbol{\theta}}[H(t) \mid S,k]]}_{\text{parameter uncertainty given scenario and A4 regime}} + \underbrace{\text{Var}_k[E_{\boldsymbol{\theta}}[H(t) \mid S,k]]}_{\text{across-A4-scenario variance given SSP}} \tag{10}$$

Here $\text{Var}_{\boldsymbol{\theta}}[H(t) \mid S,k]$ is the posterior parameter uncertainty for a fixed SSP and A4 regime, and $\text{Var}_k[\cdot]$ captures the structural disagreement among A4 scenarios about the AIS contribution. This nested structure is the correct generalization of the decomposition used by **?**; it replaces the informal additive form that appeared in an earlier draft.

5.2 Cross-component covariance

Because $H(t)$ is obtained by integrating the sum of component rates, the parameter-uncertainty term in eq. (10) inherits cross-component structure:

$$\mathrm{Var}_{\boldsymbol{\theta}}\left[\sum_i r_i(t) \,\middle|\, S, k\right] = \sum_i \mathrm{Var}_{\boldsymbol{\theta}}[r_i(t)] + C_{\mathrm{cross}}(t) \tag{11}$$

where

$$C_{\mathrm{cross}}(t) = 2\sum_{i<j} \mathrm{Cov}_{\boldsymbol{\theta}}[r_i(t),\, r_j(t)] \tag{12}$$

The quantity $C_{\mathrm{cross}}(t)$ is a sum of covariances, *not* a variance. It can be negative (anti-correlated components partially cancel) or positive (correlated amplification). The notation $\sigma^2_{\mathrm{cross}}$ used in an earlier draft was incorrect because it implied a non-negative quantity.

The dominant contributor to C_{cross} is $\mathrm{Cov}[r_{\mathrm{TWS,clim}},\, r_{\mathrm{AIS}}]$, which is negative on interannual timescales (La Niña increases land water storage while reducing CDW intrusions) and may be positive on centennial timescales (sustained El Niño–like conditions drive both TWS fluctuations and cumulative grounding-line retreat; see section 3.5).

In practice, $C_{\mathrm{cross}}(t)$ is computed directly from the MCMC posterior samples: at each draw $\boldsymbol{\theta}^{(m)}$, evaluate all component rates and compute the sample covariance across draws. No analytic approximation is required.

5.3 Horizon-dependent model inadequacy

The component-level model inadequacy terms $\sigma_{\mathrm{extra},i}$ introduced in eq. (4) are calibrated during the historical period and, absent correction, are applied unchanged during the forecast period. This is physically wrong: a model calibrated on 1900–2018 data has monotonically increasing inadequacy as it forecasts further from the calibration window, for two reasons. First, under all SSP scenarios, GMST exceeds the calibration range by ∼2040–2050, so component models operate in extrapolation with structural error that grows as the forcing departs from the observed range. Second, nonlinear feedbacks (MISI, glacier area-loss cascading, AMOC changes affecting thermosteric response) may activate at longer horizons; these processes are not captured by the calibrated parametric models and represent structural inadequacy that grows with lead time.

The scalar $\sigma_{\mathrm{extra},i}$ therefore produces overconfident forecasts at long horizons and slightly underconfident forecasts at very short horizons (where the system trajectory is tightly constrained and the parametric model is essentially interpolating).

Horizon-dependent formulation. Replace the scalar $\sigma_{\mathrm{extra},i}$ with a horizon-dependent function $\sigma_{\mathrm{extra},i}(\tau)$, where $\tau = t - t_{\mathrm{cal,end}}$ is the forecast lead time. During the calibration period ($\tau \leq 0$), the model inadequacy reduces to the calibration scalar:

$$\sigma_{\mathrm{extra},i}(\tau) = \sigma_{\mathrm{extra},i,0} \quad \text{for } \tau \leq 0 \tag{13}$$

During the forecast period ($\tau > 0$), it grows:

$$\sigma^2_{\mathrm{extra},i}(\tau) = \sigma^2_{\mathrm{extra},i,0} + g_i(\tau) \cdot \sigma^2_{\mathrm{struct},i} \tag{14}$$

where $\sigma_{\mathrm{extra},i,0}$ is the calibration-period model inadequacy (identical to the current $\sigma_{\mathrm{extra},i}$; learned from data via a HalfCauchy prior), $\sigma_{\mathrm{struct},i}$ is a structural uncertainty scale representing the maximum additional model inadequacy at long lead times (a new parameter with a component-specific

HalfCauchy prior), and $g_i(\tau)$ is a dimensionless growth function satisfying $g_i(0) = 0$ and $g_i(\tau) \to 1$ as $\tau \to \infty$, with shape determined by the physics of each component.

The total predictive model-inadequacy variance at forecast lead time τ is:

$$\sigma^2_{\text{pred}}(\tau) = \sum_{i=1}^{K} \sigma^2_{\text{extra},i}(\tau) \tag{15}$$

Component-specific growth functions. The functional form of $g_i(\tau)$ encodes how model inadequacy accumulates for each component. Table 4 specifies the growth functions and their physical rationale. The key distinctions are: (a) exponential saturation for components whose physics is well-understood and whose model error is bounded (thermosteric, SMB); (b) square-root growth for components whose structural error accumulates diffusively (glaciers); (c) linear growth for components driven by human decisions with no physical bound on future trajectory (TWS secular); (d) no growth for components where the functional form is equally valid at all times (TWS climate); (e) not directly applicable for AIS, where the A4 module provides its own horizon-dependent uncertainty.

Note on AIS. The AIS component does not use a parametric forward model during projection; the A4 module provides its own horizon-dependent uncertainty through its scenario structure. The growth function $g_i(\tau)$ applies only to components with parametric rate models. During the calibration period, the AIS component has a $\sigma_{\text{extra,AIS},0}$ that reflects the budget-residual model inadequacy; during projection, the A4 module replaces this entirely. The transition between these regimes is handled by the forecast regime structure (section 5.6).

Budget tolerance during the forecast period. The budget tolerance (eq. (4)) is modified during the forecast period. The observational uncertainty term $\sigma^2_{\text{obs,total}}(t_j)$ vanishes because there are no observations to constrain, and the component model-inadequacy terms become horizon-dependent:

$$\sigma^2_{\text{budget,forecast}}(t) = \sum_{i=1}^{K} \sigma^2_{\text{extra},i}(\tau), \qquad \tau = t - t_{\text{cal,end}} \tag{16}$$

This means the budget constraint is tighter at short forecast horizons (where model inadequacy is small, so component rates must sum closely) and progressively looser at long horizons (where model inadequacy is large, so more budget imbalance is tolerated). This is physically correct: at short horizons, the component decomposition is well-constrained; at long horizons, it is increasingly uncertain.

New parameters and Phase 1 treatment. The horizon-dependent formulation adds, per applicable component, two new parameters: $\sigma_{\text{struct},i}$ (HalfCauchy prior, component-specific scale) and $\tau_{i,\text{struct}}$ (LogNormal prior, component-specific; see table 4). These parameters are not identifiable from the calibration data alone. In Phase 1, $\tau_{i,\text{struct}}$ is fixed at the prior median; $\sigma_{\text{struct},i}$ is drawn from the prior with no likelihood update. This makes the horizon-dependent inadequacy a *prior predictive* feature: it widens the forecast uncertainty band relative to the scalar model, consistent with the physical expectation that model error grows with extrapolation distance, without requiring additional data to calibrate. In Phase 2, the hindcast cross-validation (section 6) will provide empirical constraints on both the growth-function form and the structural uncertainty scale.

5.4 Conditions for the additive approximation

The informal additive form $\sigma^2_{\text{total}} \approx \sigma^2_{\text{constrained}} + \sigma^2_{\text{scenario}} + \sigma^2_{\text{ice}} + C_{\text{cross}}$ is not wrong as an approximation, but it requires two conditions that must be stated explicitly and checked empirically.

Assumption V1 (A4/calibration independence). $\text{Var}_{\boldsymbol{\theta}}[H(t) \mid S, k]$ and $\text{Var}_k[E_{\boldsymbol{\theta}}[H(t) \mid S, k]]$ are approximately independent — i.e., the posterior parameter uncertainty does not depend strongly on which A4 scenario is assumed. This holds if the A4 module does not substantially modify the AIS rate during the calibration period, which in turn requires that the A4 scenarios are consistent with the historical budget residual. The A4/budget consistency diagnostic (Section **??**[1]) tests this condition. If A4 posterior weights differ substantially from prior weights, Assumption V1 may be violated.

Assumption V2 (parameter–scenario separability). The across-scenario variance $\text{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]$ is approximately separable from the parameter uncertainty — i.e., the temperature-to-rate mapping is approximately the same at the posterior mean as it is over the full posterior. This holds if the posterior on the rate–temperature relationship is tight enough that nonlinear parameter–temperature interactions are small.

This is a testable condition. Define the separability ratio:

$$R_{\text{sep}}(t, S) = \frac{\text{Var}_{\boldsymbol{\theta}}[r(t;\, \boldsymbol{\theta}) \mid S,\, T_S(t)]}{\text{Var}_S[r(t;\, \bar{\boldsymbol{\theta}}) \mid \bar{\boldsymbol{\theta}}]} \tag{17}$$

where $\bar{\boldsymbol{\theta}}$ is the posterior mean. The numerator measures parameter uncertainty at a fixed scenario temperature; the denominator measures scenario spread evaluated at the posterior mean.

Diagnostic (mandatory): Compute $R_{\text{sep}}(t, S)$ at $t \in \{2050, 2100, 2150\}$ for each SSP scenario. If $R_{\text{sep}} > 0.3$ for any scenario at any horizon, the interaction between parameter uncertainty and scenario spread is non-negligible, and the full nested decomposition (eqs. (9) and (10)) must be used rather than the additive approximation. Report R_{sep} values in the results.

5.5 Practical computation

In practice, the full nested decomposition is computed by Monte Carlo. For each SSP scenario S:

1. Draw M posterior parameter samples $\boldsymbol{\theta}^{(1)}, \ldots, \boldsymbol{\theta}^{(M)}$.
2. For each draw, sample an A4 scenario k from its (posterior-updated) categorical distribution.
3. Evaluate $H^{(m)}(t) = H(t;\, \boldsymbol{\theta}^{(m)}, k^{(m)}, S)$.
4. The within-scenario variance is $\widehat{\text{Var}}[H(t) \mid S] = \frac{1}{M-1} \sum_m \left(H^{(m)}(t) - \bar{H}_S(t)\right)^2$.

The across-scenario variance is computed from the scenario-specific means $\bar{H}_S(t)$. The law-of-total-variance decomposition is then exact up to Monte Carlo error, with no approximations required. The additive approximation and the R_{sep} diagnostic serve as interpretive tools, not as computational shortcuts.

As the framework matures through the staged implementation, the parameter uncertainty decreases (tighter priors from component decomposition), while the A4 contribution and C_{cross} become better characterised.

[1] Section to be added per Item 4 of the revision plan; until then, this forward reference is intentional.

5.6 Forecast regime structure

The variance decomposition reveals a natural partition of the forecast period into three regimes, distinguished by which uncertainty source dominates and by the appropriate credibility basis for the projections. The regime boundaries are not free parameters — they are emergent properties of the posterior.

Regime 1: Observational constraint dominant ($\tau < \tau_{\mathbf{obs}}$). At short lead times ($\tau \approx 0$–15 years), the system trajectory is strongly constrained by its current state, recent acceleration, and near-term committed warming. The dominant uncertainty sources are interannual variability (aleatoric; ENSO, volcanic), near-term forcing uncertainty (small; the SSP trajectories are effectively indistinguishable for ∼15 years), and parameter uncertainty in the rate–temperature relationship. Process models play a minimal role: they carry substantial initialisation errors and are outperformed by observation-constrained statistical models at these horizons. The deep-uncertainty module (A4) is negligible: the AIS contribution is small and well-constrained by the budget residual over the next decade. The credibility basis for Regime 1 projections is hindcast cross-validation (section 6): the framework must demonstrate forecast skill exceeding simple trend extrapolation at all horizons within this regime.

Regime 2: Physics-informed projection ($\tau_{\mathbf{obs}} < \tau < \tau_{\mathbf{diverge}}$). At intermediate horizons ($\tau \approx 15$–60 years), the forcing scenarios begin to diverge materially, and the component models extrapolate beyond their calibration domain. The dominant uncertainty sources are scenario uncertainty (growing), structural/model-form uncertainty (growing; captured by the horizon-dependent model inadequacy of section 5.3), and WAIS dynamics (emerging; possible early-stage MISI). The credibility basis is physical plausibility of the component models, consistency with the Regime 1 posterior at the transition point, budget closure, and process-model structural information informing priors. The hindcast validation provides partial coverage for the early part of this regime. The A4 module becomes the primary uncertainty source for the AIS contribution as the budget-residual constraint weakens.

Regime 3: Deep uncertainty dominant ($\tau > \tau_{\mathbf{diverge}}$). At long horizons ($\tau \approx 60$–200 years), the projection envelope is dominated by structural and scenario uncertainty. For well-behaved components (thermosteric, glaciers, GrIS), scenario uncertainty is the leading term. For AIS, the A4 deep-uncertainty module is dominant, with the ice-sheet contribution being the primary source of high-end risk. The credibility basis shifts from empirical validation to physical constraints (conservation laws, reservoir limits, paleoclimate analogues), internal consistency (budget closure, process understanding), and expert assessment (A4 scenario weights). There is no direct empirical validation at these horizons for the specific forcing scenarios of interest.

Transition timescales. The boundaries τ_{obs} and τ_{diverge} are approximate and scenario-dependent. Under SSP1-2.6, Regime 2 extends further because the system stays closer to the calibration domain; under SSP5-8.5, Regime 3 begins earlier because the system enters novel territory sooner. Formally:

- τ_{obs} is the *skill horizon* from the hindcast cross-validation (section 6): the lead time at which the framework's forecast skill degrades below simple extrapolation.

- τ_{diverge} is the horizon at which the across-scenario variance $\text{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]$ exceeds the within-scenario uncertainty (parameter + structural + A4) for the majority of components, computed from the variance decomposition (eqs. (9) and (10)).

These emergent timescales depend on the posterior and on the scenario set; they should be reported as scenario-conditional ranges rather than single numbers. The three-regime framing is a conceptual and communicative tool — not a mathematical partition. All three uncertainty sources contribute at all horizons; the regime labels identify which source is *dominant*. Table 5 summarises the regime structure.

5.7 Horizon-stratified uncertainty composition

The variance decomposition (eqs. (9) and (10)) combined with the horizon-dependent model inadequacy (eq. (14)) produces an uncertainty composition that changes qualitatively with forecast horizon. This is one of the most communicable results of the framework: it directly answers the question that decision-makers require — how confident should projections at 2050 vs. 2100 vs. 2150 be, and why.

Reporting horizons. The variance decomposition is evaluated at $t \in \{2030, 2050, 2075, 2100, 2150, 2200\}$ for each SSP scenario. These horizons span all three forecast regimes.

Reported quantities. At each reporting horizon t and scenario S:

1. Total projection variance $\text{Var}[H(t) \mid S]$.
2. Fractional contributions from each uncertainty source:

$$f_{\text{param}}(t, S) = \frac{E_k[\text{Var}_{\boldsymbol{\theta}}[H(t) \mid S, k]]}{\text{Var}[H(t) \mid S]} \qquad \text{(parameter uncertainty)} \tag{18}$$

$$f_{\text{A4}}(t, S) = \frac{\text{Var}_k[E_{\boldsymbol{\theta}}[H(t) \mid S, k]]}{\text{Var}[H(t) \mid S]} \qquad \text{(A4 deep uncertainty)} \tag{19}$$

$$f_{\text{struct}}(t, S) = \frac{\sum_i g_i(\tau) \cdot \sigma^2_{\text{struct},i}}{\text{Var}[H(t) \mid S]} \qquad \text{(model inadequacy)} \tag{20}$$

$$f_{\text{scenario}}(t) = \frac{\text{Var}_S[E_{\boldsymbol{\theta},k}[H(t) \mid S]]}{\text{Var}[H(t)]} \qquad \text{(scenario spread, across SSPs)} \tag{21}$$

3. Component-level fractional contributions (which physical component dominates the total variance at each horizon).
4. The forecast regime (1, 2, or 3) for each component at each horizon, based on the transition timescales from section 5.6.

Presentation. Figure 2 will present the horizon-stratified decomposition as a two-panel summary figure. Panel A shows fractional contributions of parameter, scenario, A4, and model-inadequacy uncertainty at each reporting horizon for a reference scenario (SSP2-4.5). Panel B shows fractional contributions of each physical component (thermosteric, glaciers, GrIS, AIS, TWS) to the total variance at each reporting horizon. Contributions that are prior-dominated (as opposed to data-constrained) should be visually distinguished (e.g., hatched shading) to communicate which parts of the decomposition are empirically grounded and which are informed by physical reasoning alone.

Placeholder: Horizon-stratified variance decomposition

Panel A: Stacked bar chart of uncertainty-source fractions (f_{param}, f_{A4}, f_{struct}, f_{scenario}) at each reporting horizon for SSP2-4.5. Hatched shading for prior-dominated contributions.

Panel B: Stacked bar chart of component fractions (thermosteric, glaciers, GrIS, AIS, TWS) at each reporting horizon.

To be populated when the Phase 1 posterior is available.

Figure 2: Horizon-stratified variance decomposition. The composition of total projection uncertainty changes qualitatively with forecast horizon, transitioning from parameter-dominated at short horizons through scenario-dominated at intermediate horizons to deep-uncertainty-dominated at long horizons (section 5.6). Results for two SSP-weighting schemes (uniform and plausibility-weighted) should be compared, as the scenario-uncertainty fraction is sensitive to this choice.

Sensitivity to scenario weighting. The fractional decomposition depends on the weights assigned to the SSP scenarios. Under uniform weighting, scenario uncertainty will dominate at most horizons; under plausibility-weighted schemes (e.g., downweighting SSP5-8.5), the decomposition shifts toward parameter and structural uncertainty. Results should be reported for at least two weighting schemes to make this sensitivity visible.

6 Forecast Skill Validation

The framework includes no mechanism for evaluating its own forecast skill as a function of lead time unless one is explicitly specified. Without such a mechanism, there is no empirical basis for calibrating the growth functions $g_i(\tau)$ in section 5.3, no demonstration that the framework produces reliable short-term forecasts, and no basis for comparing the framework's skill to simpler alternatives. This section defines a hindcast cross-validation protocol as a mandatory diagnostic.

6.1 Hindcast protocol

For each training cutoff year $Y \in \mathcal{Y} = \{1960, 1970, 1980, 1990, 2000\}$:

1. Fit the full joint model (eq. (7)) using only data up to year Y: GMSL observations $\mathbf{H}_{\text{obs}}(t)$ for $t \leq Y$; component datasets $\mathbf{D}_i(t)$ for $t \leq Y$ (where available — for cutoff years before the

satellite era, some component datasets are empty).

2. Generate posterior predictive distributions for $H(t)$ at forecast horizons $\tau \in \{10, 20, 30, 40, 50\}$ years, where $t = Y + \tau \leq 2018$ so that held-out observations are available for comparison.

3. Compute forecast skill metrics by comparing the posterior predictive to the held-out observations $\mathbf{H}_{\text{obs}}(Y + \tau)$.

6.2 Skill metrics

For each cutoff year Y and forecast horizon τ:

Coverage probability. The fraction of held-out observations falling within the p-percent posterior predictive interval, evaluated at $p \in \{50, 90, 95\}$:

$$\text{Coverage}_p(\tau) = \frac{1}{|\mathcal{Y}|} \sum_{Y \in \mathcal{Y}} \mathbf{1}(H_{\text{obs}}(Y + \tau) \in \text{CI}_p(Y, \tau)) \tag{22}$$

where $\text{CI}_p(Y, \tau)$ is the p-percent posterior predictive interval from the model trained on data up to year Y. Overconfidence manifests as coverage below the nominal rate; underconfidence as coverage above.

Continuous Ranked Probability Score (CRPS). The CRPS measures the full distributional forecast quality:

$$\text{CRPS}(F_{Y,\tau},\, H_{\text{obs}}(Y + \tau)) = \int_{-\infty}^{\infty} [F_{Y,\tau}(x) - \mathbf{1}(x \geq H_{\text{obs}}(Y + \tau))]^2 \, \mathrm{d}x \tag{23}$$

where $F_{Y,\tau}$ is the posterior predictive CDF from the model trained on data up to year Y, evaluated at horizon τ. Lower CRPS indicates better distributional calibration. The CRPS is a strictly proper scoring rule: it is minimised in expectation only by the true data-generating distribution.

Interval width. The width of the 90% posterior predictive interval at each horizon. What matters is not the width per se but whether the interval width grows with τ in a way that is consistent with the coverage remaining near nominal.

Skill relative to baseline. The framework is compared to two reference forecasts:

- *Linear trend extrapolation:* Fit a linear trend to $\mathbf{H}_{\text{obs}}(t)$ for $t \leq Y$ and extrapolate with Gaussian prediction interval.
- *Climatological drift:* Project the rate observed at year Y forward with no change, with prediction interval widening at $\sigma_{\text{rate}}\sqrt{\tau}$.

The framework should demonstrate positive skill (lower CRPS) relative to both baselines at all horizons up to the skill horizon τ^*.

6.3 Skill horizon

The *skill horizon* τ^* is the forecast lead time at which the framework's CRPS degrades to the level of the linear-trend baseline. Beyond τ^*, the parametric model adds no value over simple extrapolation, and the forecast should be dominated by the structural uncertainty terms $\sigma_{\text{struct},i}$ and, for AIS, by the A4 deep-uncertainty module.

The skill horizon is component-specific and forcing-scenario-dependent: models extrapolate better under mild scenarios that stay closer to the calibration range. Report τ_i^* for each component and scenario. The skill horizon τ^* for the total GMSL serves as the empirical estimate of τ_{obs}, the Regime 1/Regime 2 transition timescale in section 5.6.

6.4 Calibrating the growth functions from hindcast data

If the framework is overconfident at horizon τ (coverage below nominal for multiple cutoff years), the growth functions $g_i(\tau)$ are too slow — the model inadequacy $\sigma_{\text{extra},i}(\tau)$ is too small. Conversely, if the framework is underconfident, the growth functions are too aggressive. The hindcast residuals provide an empirical constraint:

$$g_i(\tau) \approx \frac{\hat{\sigma}^2_{\text{resid},i}(\tau) - \sigma^2_{\text{extra},i,0}}{\sigma^2_{\text{struct},i}} \tag{24}$$

where $\hat{\sigma}^2_{\text{resid},i}(\tau)$ is the empirical mean squared forecast error at horizon τ, estimated across cutoff years. In practice, the empirical signal is noisy (only 5 cutoff years, each with limited hold-out data), so a hybrid approach — using the physics-informed functional form from table 4 but calibrating its parameters from hindcast data — is likely optimal. This calibration is deferred to Phase 2; in Phase 1, the growth functions are set by the physics-informed priors alone.

Open limitations. With only 5 cutoff years, the coverage estimates have substantial sampling noise: coverage of 3/5 vs. 4/5 vs. 5/5 at the 90% level is not statistically distinguishable at conventional significance levels. The diagnostic is informative as a pattern across horizons and cutoffs, not as a point estimate. The hindcast protocol requires re-running the MCMC for each cutoff year, which is computationally feasible for Phase 1 (polynomial rate models, ∼minutes per run) but may be substantial for Phase 2 (emulator-based models). Most importantly, the hindcast does not test the framework's ability to forecast under forcing regimes that have not been observed: all held-out data are within the historical forcing envelope. The hindcast validates interpolation skill, not extrapolation skill.

7 Connection to Process Models

The framework provides a principled interface to process models that avoids treating their outputs as ground truth:

1. **Structural information → priors:** Process models (ISMIP6, GlacierMIP) inform the *shape* of component priors: the sign constraints, the expected functional form (linear vs. nonlinear), the approximate magnitude. They do *not* determine the prior location or width, which come from observations.

2. **Budget as reality check:** If a process model predicts the AIS rate to be $r_{\mathrm{AIS}} \in [0, 1.5]$ mm/yr over 2005–2020 but the budget-constrained residual implies $r_{\mathrm{AIS}} \in [0.5, 2.5]$ mm/yr, the tension is informative: either the process model is missing physics that accelerates ice loss, or the statistical decomposition is misattributing acceleration from another component.

3. **Scenario weights from process understanding:** The A4 mixture weights (Level 3) are informed by process-model ensembles and expert assessment, not by the historical calibration. This is appropriate because the deep-uncertainty scenarios represent futures that have not yet been observed.

4. **Forecast skill evaluation:** The framework provides a natural basis for evaluating process-model forecasting skill. A process model that predicts component rates inconsistent with the budget-constrained posterior (after accounting for known biases) demonstrates a lack of forecasting skill for that component.

8 Relation to Existing Work

Semi-empirical models. The Vermeer & Rahmstorf (2009) model rate $= a \cdot T + b \cdot \mathrm{d}T/\mathrm{d}t$ is a special case of the rate-and-state formulation (companion document shows $b_{\mathrm{VR09}} = -d\tau$ in the small-τ limit). The hierarchical framework generalises this by decomposing the global relationship into physically distinct components.

IPCC AR6. The AR6 assessed likely ranges are derived from process-model ensembles with expert adjustment. Our framework can incorporate these ranges as informative priors on component contributions, while using the observational budget as an independent consistency check. Where the AR6 ranges are inconsistent with the budget, the framework quantifies the tension.

Kopp et al. (2014, 2017) and related frameworks. The localised probabilistic projections of **?** use a similar component decomposition but without the global budget constraint from a rate–temperature calibration. Our framework adds this top-level constraint, which provides cross-component regularisation that purely bottom-up approaches lack.

9 Implementation Roadmap

10 Summary

The hierarchical framework provides:

1. **Self-consistency:** The rate-level budget constraint (eqs. (2) and (3)) ensures that component rates sum to the observed total at every calibration time step, at every stage of implementation.

2. **Principled uncertainty:** Bayesian priors encode physical knowledge (sign constraints, PC shrinkage, component-specific data), and the posterior honestly reflects what the observations support.

3. **Incremental refinement:** Each new component tightens the projections without invalidating previous work. The residual term absorbs unresolved components.

4. **Process-model interface:** Process models inform priors and scenario weights but are not treated as ground truth. The budget provides an independent reality check.

5. **Cross-component coupling:** The ENSO–TWS–WAIS teleconnection is explicitly represented, rather than assumed away by a diagonal error model.

6. **Transparent prior sensitivity:** The PC prior on a and the prior predictive checking ensure that the influence of the prior on projections is visible and quantifiable.

7. **Multi-timescale forecasting:** The framework explicitly recognises that the dominant uncertainty source and the appropriate credibility basis change with forecast horizon (section 5.6). Horizon-dependent model inadequacy (section 5.3) prevents overconfident long-range forecasts, and the hindcast cross-validation (section 6) provides empirical validation of short-range forecast skill. The horizon-stratified variance decomposition (section 5.7) makes the framework's own credibility as a function of lead time transparent and quantifiable.

The framework is designed not to produce a single "best estimate" of future SLR, but to provide a *defensible probabilistic envelope* that honestly reflects what is known, what is uncertain, and what is deeply uncertain — and to do so in a way that is formally self-consistent across scales, components, time horizons, and methodological approaches.

Table 1: Complete node set of the framework DAG. Node types: **F** = exogenous forcing (Level 0), **M** = mediating variable, **C** = component rate (Level 2), **G** = global constraint (Level 1), **D** = deep uncertainty (Level 3), **O** = observed data.

Node	Symbol	Type	Description
SSP scenario	SSP	F	Shared Socioeconomic Pathway label; indexes the forcing trajectory. Discrete.
Radiative forcing	$F(t)$	F	Net anthropogenic radiative forcing ($\mathrm{W\,m^{-2}}$), conditioned on SSP.
Global mean temperature	$T_{\mathrm{GMST}}(t)$	F	Global mean surface temperature anomaly relative to pre-industrial.
North Atlantic SST proxy	$T_{\mathrm{NA}}(t)$	M	Subsurface ocean temperature relevant to GrIS marine-terminating outlet glaciers. Mediates GMST $\rightarrow$ GrIS discharge.
ENSO index	$E(t)$	M	El Niño–Southern Oscillation state (e.g., Niño 3.4 index or SOI). Mediates multiple cross-component couplings.
CDW temperature proxy	$T_{\mathrm{CDW}}(t)$	M	Circumpolar Deep Water temperature on the Antarctic continental shelf. Mediates ENSO $\rightarrow$ AIS forcing.
Thermosteric rate	$r_{\mathrm{th}}(t)$	C	Ocean thermal expansion rate ($\mathrm{mm\,yr^{-1}}$).
Glacier rate	$r_{\mathrm{gl}}(t)$	C	Glaciers and ice caps mass-loss rate ($\mathrm{mm\,yr^{-1}}$ SLE).
GrIS SMB rate	$r_{\mathrm{SMB}}(t)$	C	Greenland surface mass balance rate.
GrIS discharge rate	$r_{\mathrm{dis}}(t)$	C	Greenland ice discharge rate (marine-terminating outlet glaciers).
AIS rate	$r_{\mathrm{AIS}}(t)$	C	Antarctic Ice Sheet mass-loss rate. Budget residual during calibration; A4 module during projection.
TWS secular rate	$r^{\mathrm{s}}_{\mathrm{TWS}}(t)$	C	Terrestrial water storage secular rate (dams, groundwater).
TWS climate rate	$r^{\mathrm{c}}_{\mathrm{TWS}}(t)$	C	Terrestrial water storage climate-driven rate (ENSO-mediated).
Total rate	$r_{\mathrm{total}}(t)$	G	Deterministic sum of component rates (eq. (2)). Constrained via soft likelihood (eq. (3)).
GMSL	$H_{\mathrm{GMSL}}(t)$	G	Global mean sea level, obtained by integrating r_{total}.
A4 scenario index	k	D	Discrete index over AIS deep-uncertainty scenarios (S1–S4).
Scenario weights	w_k	D	Prior mixture weights from expert assessment; updated by budget consistency.
GMSL observations	$\mathbf{H}_{\mathrm{obs}}$	O	Frederikse et al. (2020) reconstruction, 1900–2018.
Component datasets	$\mathbf{D}_{\mathrm{th}}, \mathbf{D}_{\mathrm{gl}}, \mathbf{D}_{\mathrm{GrIS}}, \mathbf{D}_{\mathrm{AIS}}, \mathbf{D}_{\mathrm{TWS}}$	O	Component-specific observational records (Argo, GlaMBIE, IMBIE, GRACE, etc.).

Table 2: Intervention targets and their DAG operations. Each intervention corresponds to a do-operation on a specific node, with the specified incoming edges severed.

Intervention	DAG operation	Downstream effects
Stabilise Thwaites grounding line	$do(r_{\mathrm{AIS}} = f_{\mathrm{stab}}(t))$ where f_{stab} specifies the stabilised trajectory. Severs $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$ and $k \to r_{\mathrm{AIS}}$.	r_{total}, H_{GMSL} change; all other component rates unaffected.
Reduce basal melt via MCB	$do(T_{\mathrm{CDW}} = T_{\mathrm{CDW}} - \Delta T_{\mathrm{MCB}})$. Severs $E(t) \to T_{\mathrm{CDW}}$; replaces with modified mechanism.	r_{AIS} responds through the $T_{\mathrm{CDW}} \to r_{\mathrm{AIS}}$ edge; cascades to r_{total}, H_{GMSL}. Other components unaffected unless MCB also modifies $E(t)$.
Stratospheric aerosol injection	$do(F(t) = F_{\mathrm{SAI}}(t))$. Severs SSP $\to F(t)$; replaces with modified forcing trajectory.	T_{GMST} changes; GMST-forced components (r_{th}, r_{gl}, r_{SMB}) and r_{dis} (via T_{NA}) respond directly. **Phase 1**: with ENSO exogenous, SAI does *not* propagate to r_{AIS} or $r^{\mathrm{c}}_{\mathrm{TWS}}$ via the ENSO pathway. **Phase 2**: with $T_{\mathrm{GMST}} \to E(t)$ active, SAI additionally propagates through ENSO to $r^{\mathrm{c}}_{\mathrm{TWS}}$ and to r_{AIS} via $E(t) \to T_{\mathrm{CDW}}$.
Carbon dioxide removal	$do(F(t) = F_{\mathrm{CDR}}(t))$. Same structural operation as SAI but different temporal profile.	Same propagation as SAI (including the Phase 1/Phase 2 distinction regarding the ENSO pathway); slower onset, persistent effect.
Combined CDR + MCB	$do(F(t) = F_{\mathrm{CDR}}(t))$, $do(T_{\mathrm{CDW}} = g(T_{\mathrm{CDW}}, \mathrm{MCB}))$. Simultaneous intervention on two nodes.	Global forcing reduction (via CDR) plus targeted Antarctic shelf cooling (via MCB). The DAG structure permits computing the joint effect because the two intervention targets have no common descendant that is also an ancestor of the other.

Table 3: Component rate models and forcing variables. Each component defines its own functional form with its own physically appropriate forcing; no shared polynomial-in-T parameterisation is imposed.

Component	Rate model	Forcing variable(s)	Justification
Thermosteric	$b_{\mathrm{th}}T + c_{\mathrm{th}}$	GMST	Direct thermodynamic response
Glaciers	$a_{\mathrm{gl}}T^2 + b_{\mathrm{gl}}T + c_{\mathrm{gl}}$	GMST	Melt + area-loss feedback
GrIS (SMB)	$b_{\mathrm{SMB}}T + c_{\mathrm{SMB}}$	GMST	Surface energy balance
GrIS (discharge)	$\beta_{\mathrm{dis}}T_{\mathrm{NA}} + c_{\mathrm{dis}}$	North Atlantic SST	Ocean-forced outlet dynamics
AIS	Budget residual (calibration); A4 module (projection)	CDW proxy	Ocean-forced dynamics
TWS (secular)	c_{TWS}	Dam/GW records	No temperature dependence
TWS (climate)	$\gamma_{\mathrm{TWS}} \cdot E(t)$	ENSO index	Precipitation-mediated

Table 4: Component-specific growth functions $g_i(\tau)$ for the horizon-dependent model inadequacy. The characteristic timescale $\tau_{i,\text{struct}}$ governs the rate of growth; $\sigma_{\text{struct},i}$ is the asymptotic structural uncertainty scale. In Phase 1, $\tau_{i,\text{struct}}$ is fixed at the prior median; $\sigma_{\text{struct},i}$ is drawn from the prior with no likelihood update.

Component	$g_i(\tau)$	Rationale	$\tau_{i,\text{struct}}$ prior
Thermosteric	$1 - \exp(-\tau/\tau_{\text{th,struct}})$	Slow growth; physics well-understood; bounded thermal expansion coefficient variability.	$\text{LogN}(\ln 150, 0.5)$
Glaciers	$\min\big(1, [\tau/\tau_{\text{gl,struct}}]^{1/2}\big)$	Moderate growth; area-loss feedback nonlinear in intermediate regime; volume depletion provides natural ceiling.	$\text{LogN}(\ln 70, 0.5)$
GrIS (SMB)	$1 - \exp(-\tau/\tau_{\text{SMB,struct}})$	Similar to thermosteric; well-constrained by temperature; risk is nonlinear albedo feedback.	$\text{LogN}(\ln 100, 0.5)$
GrIS (discharge)	$1 - \exp(-\tau/\tau_{\text{dis,struct}})$	Faster growth; discharge depends on fjord geometry and ice-ocean interaction, poorly constrained by historical data.	$\text{LogN}(\ln 40, 0.5)$
AIS	Not applicable — A4 deep-uncertainty module provides horizon-dependent uncertainty during projectio		
TWS (secular)	$\tau/\tau_{\text{TWS,struct}}$	Linear growth; anthropogenic parameter trajectory (dams, groundwater) depends on human decisions with no physical bound.	Set s.t. $\sigma_{\text{struct,TWS}}\sqrt{g(80)} \approx 0.5$ m
TWS (climate)	0 (no growth)	Model $\gamma_{\text{TWS}} \cdot E(t)$ is equally valid at all times; uncertainty is in the ENSO trajectory, not model form.	N/A

Table 5: Summary of the three forecast regimes. The regime boundaries are emergent properties of the posterior, not preset values; approximate horizons are given for orientation.

Property	Regime 1 ($\tau < \tau_{\text{obs}}$)	Regime 2 ($\tau_{\text{obs}} < \tau < \tau_{\text{diverge}}$)	Regime 3 ($\tau > \tau_{\text{diverge}}$)
Approx. horizon	0–15 yr	15–60 yr	60–200 yr
Dominant methodology	Rate-and-state model; observation-constrained	Component parametric/emulator models; physics-informed	Process-model-informed; A4 deep-uncertainty
Dominant uncertainty	Parameter + aleatoric	Scenario + structural	Deep uncertainty + scenario
Credibility basis	Hindcast cross-validation	Physical plausibility; budget closure	Conservation laws; paleo constraints; expert assessment
A4 role	Negligible	Growing	Dominant (AIS)
Model inadequacy	$\sigma_{\text{extra},i,0}$ (calibration value)	$g_i(\tau)$ growing	$g_i(\tau) \to 1$; saturated

Table 6: Staged implementation of the hierarchical framework.

Stage	Component added	Key data	Status
0	Global GMSL–GMST constraint	Frederikse GMSL, Berkeley T	Complete
	+ PC prior on a, rate-and-state		Complete
	+ A4 deep-uncertainty module	IPCC, expert assessment	Complete
1	Thermosteric decomposition	Argo, ocean reanalysis	Planned
2	Glaciers + Greenland	RGI, GRACE, SMB models	Planned
3	TWS (secular + ENSO-driven)	GRACE, dam/GW records, SOI	Planned
4	AIS + ENSO–CDW coupling	GRACE, altimetry, ocean T	Planned
Cross-cutting capabilities			
–	Hindcast cross-validation	Held-out GMSL observations	Phase 1
–	Sequential update interface (design)	–	Phase 1
–	Sequential update (SMC implementation)	Annual observation updates	Phase 2
–	Horizon-dependent model averaging	Multiple model classes	Phase 2